Conversion of Camphene to Camphor: Day 3

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CHEM 2081 Section 003

Introduction:

The purpose of this experiment was to synthesize borneol to camphor. The organic product achieved in the lab was the final collection of this multistep synthesis of converting camphene to camphor.

Materials and Methods:

In the long-neck round-bottom flask, 0.3g of borneol and 0.3mL of glacial acetic acid was added. In addition, 1mL of acetone was added with a boiling chip. A reflux was assembled with the apparatus resting in a water bath with a temperature between 40°C-50°C. Once dissolved, 6 mL of aqueous sodium hypochloride solution was added to the mixture. This was done dropwise over the span of 35 minutes and after each addition the mixture was swirled. After the final bleach addition, the reaction was let to run an additional 15 minutes. The mixture was let to cool to room temperature and the camphor was then extracted with two portions of 2mL of methylene chloride in a centrifuge tube. The combined methylene chloride layers were neutralized with around 2 mL of saturated sodium chloride. The organic solution was then washed with 2 mL 5% sodium bisulfite and 2 mL of water. The organic solution was left to dry of anhydrous sodium sulfate. The DCM was placed over the water bath and let to evaporate. The dried solution was weighed to determine percent yield.

Results:

Watch Glass Weight	33.960 g
Watch Glass Weight with Sample	34.243 g
Sample Weight	0.283 g

Calculations:

Theoretical Yield:

(0.305 g borneol) X (1.01 g/ml) = 0.308 g borneol

(0.283g of camphor) X (152.23 g/mol) = 0.0019 mol camphor

(0.308g borneol) X (154.25 g/mol) = 0.002 mol borneol

(0.002 mol borneol) X (152.23 g/mol) = 0.304 g camphor

Percent Yield:

Actual Yield = 0.283 g of camphor

((0.283 g) / (0.304 g)) X 100 = **93.09%**

Discussion:

In this experiment, borneol was converted to camphor using glacial acetic acid and acetone. There were two sources of error, which includes solvent loss through evaporation and side products during the water bath heating. First, solvent loss through evaporation is a potential source of error if the temperature of the water became too hot for the apparatus. A solution for this would be to adjust the rheostat to a lower temperature or add cooler water to the water bath to bring down the temperature. Next, a possible side product may have formed during heating of the mixture if the reaction was superheated. Again, a solution to this would be to adjust the rheostat or add cold water so the water bath temperature didn't get too hot. In addition the actual yield is lower than the theoretical yield. This could be because of the loss of product in the experiment.

References:

H McNeil. Reduction of a Ketone. *Chem 2081 Organic Chemistry II Lab Manual.* Macmillan Learning Curriculum Solution: Plymouth, 2023, (p. 33-38).

Exp. No.	Experiment/Subject	Date
	Camphene → camphor ~~***~~ D3	11/14
Name	Lab Partner	Locker/ Desk No.	Course & Section No.
Shelby Ludwig			

Procedure:

1. In the long neck round bottom flask add racemic 0.3g of your borneol product w/ 0.3mL of glacial acetic acid. Dissolve mixture in 1mL of acetone, add a boiling chip
2. Just like in day 2, assemble a reflux and place apparatus in a water bath between 40-50°C, if borneol doesn't dissolve at this temp add 0.3mL of acetone to mixture
3. once all reagents are dissolved add 1mL of aqueous sodium hypochlorite solution to the top opening of the condenser and should be placed dropwise over 35mins after each addition, swirl the mixture to ensure mixing of chemicals, the reaction must be allowed to continue for 15mins after last bleach addition → slowly
4. next, let the mixture cool to room temp and extract camphor in 2 portions of methylene chloride, use a centrifuge tube
6. neutralize combined methylene chloride layers w ~2mL of saturated sodium bicarbonate → organic isolated layer not both layers
6. next wash the organic solution if necessary w 2mL 5% sodium bisulfite and 2mL of water
1. dry the organic ~~layer~~ solution over anhydrous sodium sulfate. free rolling crystals
5. evaporate the DCM over a water bath w temp not exceeding 30°C, weigh collected crystals.

DCM: 1.33g/mL bottom
organics will sit at bottom

if it keeps bubbly add more
↳ red litmus paper

bottom

0.283g

Signature

Lab Performance

Exp 6 D3

Shelby Ludwig
Name: *Section:* *Room:* *Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves.	○ ○ ○ ○ ~~○~~ (5pts)
Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks.	○ ○ ○ ○ ~~○~~ (5pts)
Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts).	○(1pt) ○(1pt) ○(1pt) ○ (1pt) ~~○ (5pts)~~
Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts).	○(1pt) ○ (5pts) ~~○~~ (5pts)
Working independently or equal partner participation during the entire lab.	○ ○ ○ ○ ~~○~~ (5pts)
Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace.	○ ○ ○ ○ ~~○~~ (5pts)
Maintain clean and organized personal space.	○ ○ ○ ○ ~~○~~ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* [ ]